

Leon D. Lotter

Physician Scientist · Dr. med.

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Positions

Sep 2023 – Ongoing	Research associate (100%) <i>Institute of Systems Neuroscience, University Hospital Düsseldorf, Germany</i> <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> Working group: Biomarker Development (Jürgen Dukart)
Sep 2022 – Aug 2023	Doctoral fellow (stipend) <i>Clinician Scientist Program, Max Planck School of Cognition, Germany</i> - Graduate courses in neuroscience - Lab rotation: INM-1: Structural and Functional Organisation of the Brain, Research Centre Jülich (Katrin Amunts Timo Dickscheid) - Lab rotation: Max Planck Institute of Psychiatry, Munich (Elisabeth Binder Victor Spoormaker)
Apr 2022 – Aug 2022	Research assistant (65%) <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> Project: Linking cortical thickness development to multilevel brain systems
May 2019 – Jan 2022	Student research assistant (20%) <i>Child Neuropsychology, University Hospital RWTH Aachen, Germany</i> Diverse projects involving neuroimaging, behavioral data analysis, and visualization
Sep 2013 – Aug 2014	Voluntary service <i>Samuha Samarthya, India Service Civil International, Germany</i> - weltwärts program, German government-funded - Creating barrier-free environments for people with disabilities in rural South India

Education

Sep 2022 – Ongoing	PhD in neuroscience <i>Clinician Scientist Program, Max Planck School of Cognition, Germany</i> - Cumulative thesis on the neurobiological basis of human brain imaging - Supervision: Jürgen Dukart Simon Eickhoff Julian Koenig Svenja Caspers - Completion planned in 2027
Dec 2017 – Sep 2022	Doctor of medicine (Dr. med.) <i>Child Neuropsychology, University Hospital RWTH Aachen, Germany</i> <i>INM-7: Brain and Behaviour, Research Centre Jülich, Germany</i> - Thesis: Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa - Supervision: Kerstin Konrad Jürgen Dukart Jochen Seitz - Grade: <i>summa cum laude</i> (with distinction)
Oct 2014 – Nov 2021	Medical studies (state examination) <i>RWTH Aachen University, Germany</i> - Elective subject: Clinical neuroscience - Clinical internships: Child and adolescent psychiatry (6 mos) Adult psychiatry (1 mo) Internal medicine (4 mos) General surgery (4 mos)

Grants & Research Funding

2025	Co-investigator: DFG Research Grant • Project: Linking brain functional trajectories in neurodevelopmental disorders to underlying neurotransmitter systems • Substantial contribution to conception and writing • Direct supervision of two researchers employed on the grant, until 2028
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Awards & Scholarships

April 2025	Heine Research Academies (HeRA) travel grant of the Heinrich Heine University • To present at SOBP 2025 in Toronto, Canada
Dec 2024	DGPPN 2024 "Best Free Talks Abstract" award • See conference section, awarded with 500€
Oct 2023	"Borchers Badge" for excellent dissertations at RWTH Aachen University • For the medical doctoral thesis "Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa" passed with distinction in 2022
Jul 2023	German Academic Exchange Service (DAAD) travel grant • To present at OHBM 2023 in Montréal, Canada
Jun 2023	72nd Lindau Nobel Laureate Meeting (Physiology and Medicine) Lindau, Germany • Participation as "Young Scientist", supported by Research Centre Jülich
Sep 2022	Doctoral fellowship of the Max Planck School of Cognition • Clinician Scientist Program

Supervision & Teaching

Postdoctoral	Co-supervisor: Rachel Bamberger, Research Centre Jülich (since Jan 2026; DFG grant)
Doctoral	Co-supervisor: Naimi Quitschau, University Hospital Leipzig (since Jan 2026; DFG grant)
Bachelor's	Co-supervisor: Laura Buchloh, University of Regensburg (completed 2026) Co-supervisor: Alfred Dawalibi, Aachen University of Applied Sciences (completed 2026)
Supervisor training	Workshop "Leading & Coaching Doctoral Researchers", JuDocS qualification program for supervisors, Research Centre Jülich (Mar 2026)
Teaching	Substitute lectures, Translational Neuroscience (MSc), Heinrich Heine University Düsseldorf

Academic Contributions & Personal Skills

Software Tools	NiSpace Neuroimaging Spatial Colocalization Environment mapconn Neurophysiologically Enriched Organization of Functional Connectivity (NEOFC) ABAnnotate Gene-category enrichment analysis of brain maps using Allen Human Brain Atlas expression data
Peer Reviews	<i>Journal of the American Academy of Child and Adolescent Psychiatry</i> <i>Neuropsychopharmacology</i> <i>Translational Psychiatry</i> <i>Schizophrenia Bulletin</i> <i>Neuroimage</i> <i>Neuroimage Clinical</i> <i>Cortex</i> <i>Frontiers in Human Neuroscience</i> <i>BMC Psychiatry</i> <i>BMJ Open</i> <i>European Journal of Neuroscience</i>
Extracurricular Activities	Cohort Student Representative of the Max Planck School of Cognition (since Sep 2022) Doctoral Student Representative of the INM-7, Research Centre Jülich (Oct 2022 – Dec 2024) Co-organizer of the INM-7 Journal Club , Research Centre Jülich (since Oct 2023)
Programming Languages	Python since 2021 (<i>example</i>) R since 2020 Matlab since 2019 (<i>example</i>) German native speaker English professional proficiency

Conference Presentations & Workshop Participation

Jun 2026	Organization for Human Brain Mapping (OHBM) Annual Meeting Bordeaux, France Poster: Linking human brain functional connectivity to underlying neurotransmitter systems
April 2025	Society of Biological Psychiatry (SOBP) Toronto, Canada Talk: Molecular imaging-informed biomarkers link typical and atypical functional brain synchronization to underlying neurotransmission
Dec 2024	Congress of the German Association for Psychiatry and Neurology (DGPPN) Berlin, Germany Talk: Evidence of dopaminergic modulation of resting state functional connectivity alterations in psychosis
Aug 2024	Cologne Summer School in Biological Psychiatry Cologne, Germany Participation in lectures and practical exercises on state-of-the-art methods in biological psychiatry
Jun 2024	Organization for Human Brain Mapping (OHBM) Annual Meeting Seoul, South Korea - Poster: NiSpace – Neuroimaging Spatial Colocalization Environment - Poster: Neurotransmitter systems explain lifespan changes of human resting-state brain activity
Jul 2023	Organization for Human Brain Mapping (OHBM) Annual Meeting Montréal, Canada Poster: Human cortex development is shaped by molecular and cellular brain systems
Mar 2023	Minerva Symposium: Interactive Brains - From Methods to Applications Tel Aviv, Israel Invited talk: Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion
Mar 2023	Meeting of the German Association for Child and Adolescent Psychiatry (DGKJP) Essen, Germany Poster: Linking cortical thickness development to molecular and cellular brain systems

Publications & Preprints

Preprint	Lotter, L. D. , Shafiei, G., Larabi, D., Koushik, A., Dipasquale, O., Mehta, M., Cercignani, M., Sethi, A., Harrison, N., Holiga, Š., Umbricht, D., Yakushev, I., Muthukumaraswamy, S., Forsyth, A., Hipp, J. F., Misić, B., Caspers, S., Koenig, J., Patil, K. R., Paquola, C., Eickhoff, S. B., and Dukart, J. “Linking human brain functional connectivity to underlying neurotransmitter systems”. In: <i>bioRxiv</i> . Afshani, M., Saberi, A., Reimann, G. M., Chu, C., Elmenhorst, D., Lotter, L. D. , Valk, S. L., Dukart, J., Genon, S., Müller, V. I., Eickhoff, S. B., and Tahmasian, M. “Convergent connectivity of sleep disorders and its neurotransmitter and cell enrichment correlates”. In: <i>medRxiv</i> . Bamberger, R., Kuhles, G., Lotter, L. D. , Dukart, J., Konrad, K., Günther, T., Siniatchkin, M., Fuchs, M., and von Polier, G. “Speech-based markers in paediatric ADHD: a longitudinal case-control study of voice features and medication effects”. In: <i>medRxiv</i> .
2026	Bamberger, R., Lotter, L. D. , Nieto, N., Poulain, T., Körner, A., Kiess, W., Fuchs, M., and von Polier, G. “Voices of change: Associations between vocal markers and symptoms of ADHD - Findings from the LIFE Child Study”. In: <i>European Child & Adolescent Psychiatry</i> . Dukart, J., Lotter, L. D. , Paquola, C., Eickhoff, S. B., and Schilbach, L. “The many-to-many problem of endophenotypes in psychiatry - a biological perspective”. In: <i>Molecular Psychiatry</i> . Keller, L., Lotter, L. D. , Eickhoff, C. R., Eickhoff, S. B., Otten, K., Herpertz-Dahlmann, B., and Seitz, J. “Structural brain alterations in anorexia nervosa: A global brain volume and anatomical likelihood estimation (ALE) meta-analysis combined with a functional decoding approach”. In: <i>NeuroImage: Clinical</i> .
2025	Lotter, L. D. and Dukart, J. “Methodological Considerations: Integrating Measures Across Assessment Modalities”. In: <i>Neurobehavioral Individual Differences: A Transdisciplinary Approach To Advancing Clinical Science</i> . Ed. by R. D. Latzman and C. J. Patrick. Springer Nature Switzerland. Saberi, A., Wischniewski, K. J., Jung, K., Lotter, L. D. , ... IMAGEN-Consortium, Paus, T., Dukart, J., Bernhardt, B. C., Popovych, O. V., Eickhoff, S. B., and Valk, S. L. “Adolescent maturation of cortical excitation-inhibition ratio based on individualized biophysical network modeling”. In: <i>Science Advances</i> .

- 2024 | Dukart, J., **Lotter, L. D.**, and Eickhoff, S. B. "Moving towards precision psychiatry: the hard nut of depression". In: *Signal Transduction and Targeted Therapy*.
- Lotter, L. D.**, Saberi, A., Hansen, J. Y., Misic, B., Paquola, C., ... IMAGEN-Consortium, Nees, F., Banaschewski, T., Eickhoff, S. B., and Dukart, J. "Regional patterns of human cortex development correlate with underlying neurobiology". In: *Nature Communications*.
- Lotter, L. D.**, Nehls, S., Losse, E., Dukart, J., and Chechko, N. "Temporal dissociation between local and global functional adaptations of the maternal brain to childbirth: A longitudinal assessment". In: *Neuropsychopharmacology*.
- Kasper, J., Caspers, S., **Lotter, L. D.**, Hoffstaedter, F., Eickhoff, S. B., and Dukart, J. "Resting state changes in aging and Parkinson's disease are shaped by underlying neurotransmission – a normative modeling study". In: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
- 2023 | Schloesser*, L., **Lotter*, L. D.**, Offermann, J., Borucki, K., Biemann, R., Seitz, J., Konrad, K., and Herpertz-Dahlmann, B. "Sex-dependent clinical presentation, body image, and endocrine status in long-term remitted anorexia nervosa". In: *European Eating Disorders Review*.
- Corneille, O., Havemann, J., Henderson, E. L., IJzerman, H., Hussey, I., Orban de Xivry, J.-J., Jussim, L., Holmes, N. P., Pilacinski, A., Beffara, B., Carroll, H., Outa, N. O., Lush, P., and **Lotter, L. D.** "Beware 'persuasive communication devices' when writing and reading scientific articles". In: *eLife*.
- Lotter, L. D.**, Kohl, S. H., Gerloff, C., Bell, L., Niephaus, A., Kruppa, J. A., Dukart, J., Schulte-Rüther, M., Reindl, V., and Konrad, K. "Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion". In: *Neuroscience and Biobehavioral Reviews*.
- 2021 | **Lotter, L. D.**, von Polier, G., Offermann, J., Buettgen, K., Stanetzky, L., Eickhoff, S. B., Konrad, K., Seitz*, J., and Dukart*, J. "Recovery-associated resting-state activity and connectivity alterations in anorexia nervosa". In: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
- 2020 | Pankert, K., Pankert, A., **Lotter, L. D.**, Herpertz-Dahlmann, B., and Konrad, K. "Autism spectrum symptoms in children with congenital blindness". In: *Zeitschrift für Kinder- und Jugendpsychiatrie und Psychotherapie*.

* Equal contributions